

CASE STUDY / CLOUD-MANAGED CAMPUS • CORE SWITCHING • WIRED & WIRELESS

One campus. A connected network. Central visibility.

NETAFIM | Cisco Meraki

Network design, configuration & implementation

A technical delivery blueprint for replacing existing switches, connecting the campus over fibre and managing wired and wireless services through Meraki Dashboard.

MS390 core • MS350 PoE access • MR46 / MR70 Wi-Fi

2

CORE SWITCHES

6

POE ACCESS SWITCHES

35

WIRELESS ACCESS POINTS

3 years

ORDERED LICENSE TERM

Project context | Customer requirement and ordered scope

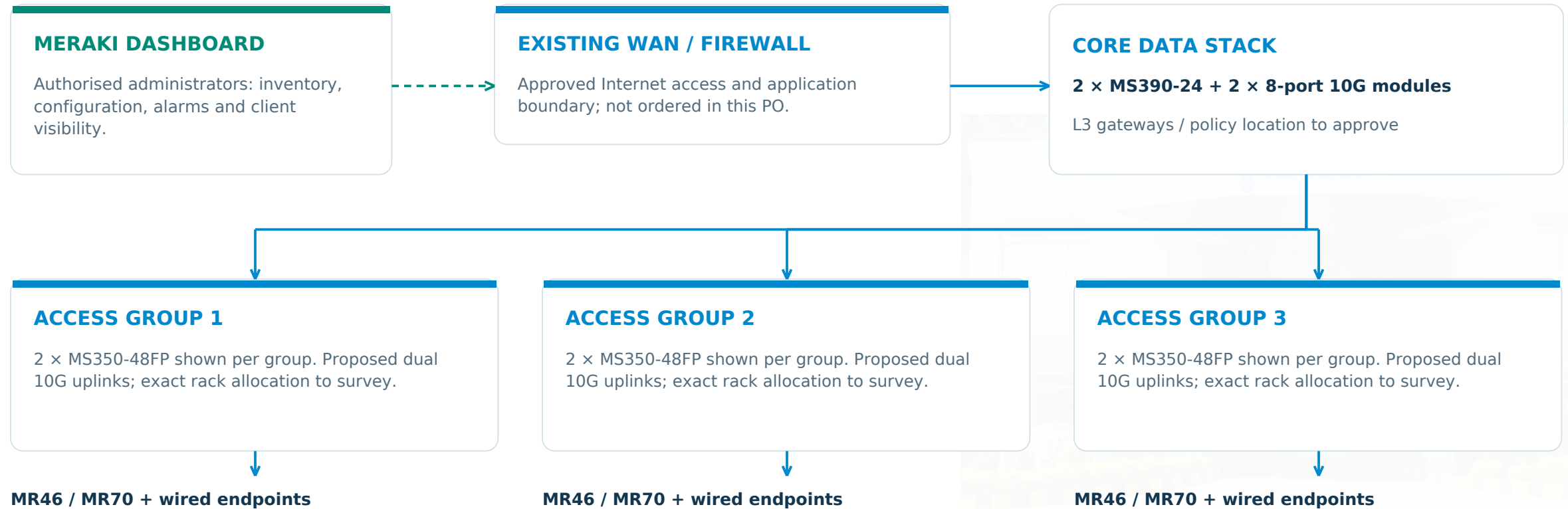
The order calls for removal of existing switches and installation of the new network with cable dressing, testing, programming and documentation. The design translates that scope into a core-and-access architecture, segmented user services and centrally managed Wi-Fi. The two supplied PDFs contain the same order and are counted once.

Layer	Ordered equipment / quantity	Delivery interpretation
CORE	2 × MS390-24-HW; 2 × MA-MOD-8X10G	Two core switches, each with one 8-port 10G uplink module.
STACK	2 × 120G / 50cm data-stack cables; 2 × 30cm Power-Stack cables	Core data-stack ring and power-stack design; final topology / power budget to verify.
ACCESS	6 × MS350-48FP-HW, 48 GigE / 740W PoE	288 ordered copper access ports; availability after uplinks / assignments must be mapped.
WIRELESS	30 × MR46 indoor; 5 × MR70	35 APs. MR46 is Wi-Fi 6; MR70 is Wi-Fi 5. Placement requires an RF survey.
LICENSES	2 core + 6 access switch licenses; 35 MR licenses, 3-year lines	Check current entitlement separately; the 2020 PO does not prove active licensing today.
SERVICES	1 installation / programming / documentation package	Switch replacement, dressing, testing and implementation documentation are in the order.

03 Architecture | 04 Stack / fibre | 05 VLANs | 06 Wireless | 07 Configuration | 08 Cutover | 09 Acceptance & support

Case study basis: ordered scope plus technical delivery design. No as-built IP plan, floor plan, Dashboard export or signed acceptance report was supplied.

Architecture | Core, access and cloud management

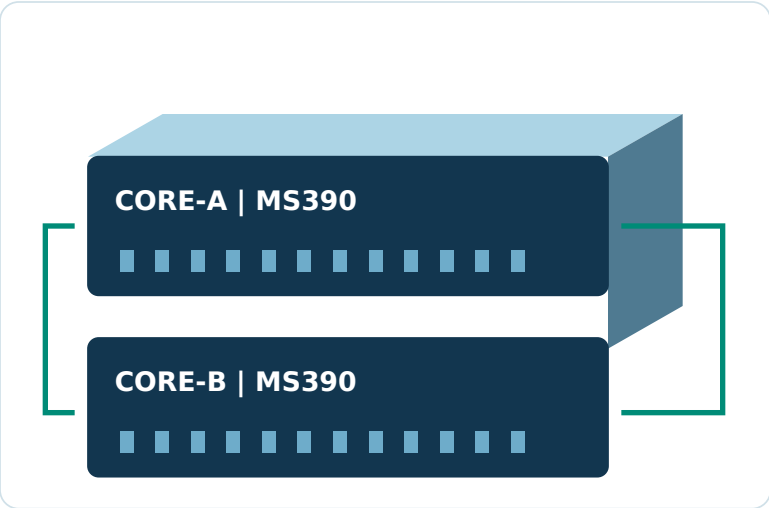


Access groups are drawing labels, not recorded site locations. Six independent access switches are the baseline; access stacking is an option to verify.

SOLID BLUE: local data path | DASHED GREEN: secure cloud management through the WAN. User application traffic does not transit Dashboard.

Physical design | Stack, optics and rack connectivity

Use the ordered modules and media to create a labelled rack-to-rack link schedule. Confirm fibre type, distance, connector path and switch compatibility before patching. Record both endpoints and actual port IDs; the example below is a design worksheet.



2 data-stack cables: closed ring

Ordered media	Qty	Engineering check
10G LR single-mode optics	20	Match SMF route, connector adaptation and receive-power limits.
10G SR multimode optics	4	Match MMF grade / distance; verify both endpoint compatibility.
Mellanox SR modules	2	Server / NIC counterpart is not identified in the PO.
10G twinax 3m + 40G QSFP 3m	2 + 2	40G cables require compatible 40G ports; the ordered core module has 10G ports.
Copper + fibre patch cords	200 + 22	PO descriptions contain colour / media inconsistencies; reconcile delivery labels.

Six dual-uplink access switches would use 12 × 10G links and 24 LR optics if every link uses LR at both ends; the PO orders 20 LR optics.

Therefore allocate LR / SR / DAC by verified route, or add approved optics. Do not assume the ordered LR count completes that proposed topology.

Segmentation | VLANs, gateways and port profiles

Example VLAN	Service boundary	Configuration / verification
10 MANAGEMENT	Switch / AP administration	Approved management IP, gateway, DNS and cloud reachability; restrict admin sources.
20 CORPORATE	Corporate wired / Wi-Fi clients	Access ports untagged VLAN 20; corporate SSID bridged to VLAN 20 or approved dynamic assignment.
30 GUEST	Visitor wireless access	Internet-only policy; deny internal networks and apply suitable isolation / bandwidth limits.
40 SERVER	Existing business services	Allow only approved client-to-server ports; record gateway / return routes.
50 RESTRICTED	Approved plant / device services	Use only if required by customer; explicit allow-list boundary to corporate services.

CORE & UPLINK PROFILE

Choose one gateway location: core SVIs or existing firewall. Align permitted VLANs, native VLAN and LACP settings at both ends; place the STP root deliberately.

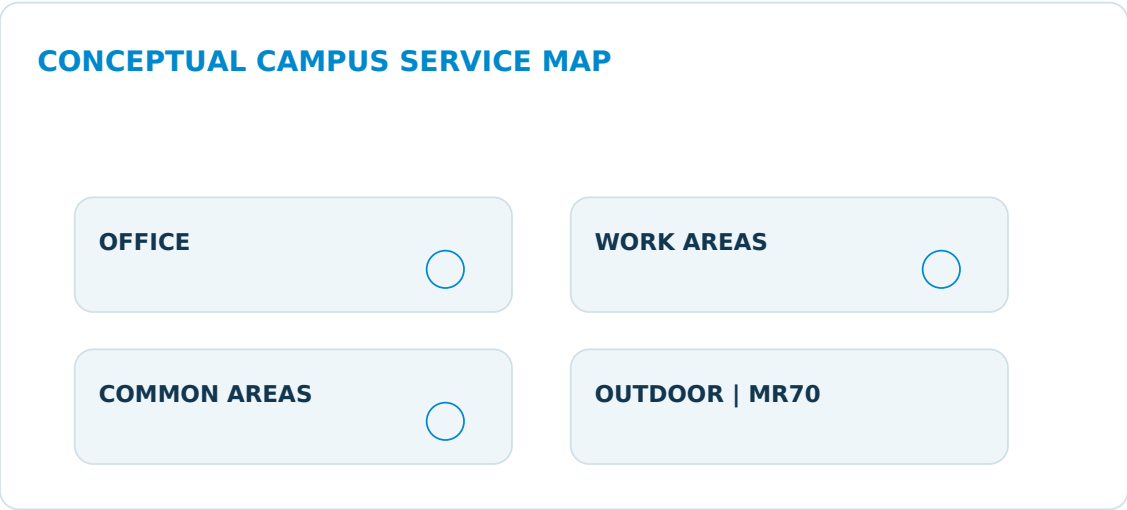
EDGE & AP PROFILE

User port: approved access VLAN. AP port: PoE plus approved AP-management VLAN and SSID VLAN tags. Disable unused ports; apply compatible edge protections.

Example VLANs are not installed values. Guest isolation needs an enforced policy at every relevant boundary, not merely a separate SSID.

Wireless design | Indoor coverage and outdoor access

Separate indoor and outdoor deployment decisions. MR46 units provide the ordered indoor Wi-Fi 6 layer; MR70 units provide Wi-Fi 5 access for approved exterior areas. The sketch is a conceptual service map, not a floor plan or measured coverage plot.



MR46: 30 indoor | MR70: 5 exterior

Design element	Implementation detail
SURVEY / LOCATION	Record floor plans, mounting height, attenuation, user density and cable routes; approve all 35 locations.
RF PROFILES	Set regulatory domain; plan channel width / power and verify interference. Tune using the post-installation survey.
SSID / AUTH	Corporate: customer-approved enterprise authentication where identity services exist. Guest: isolated visitor policy.
PoE / CAPACITY	MR46 needs 802.3at. Check per-port negotiation and switch budget; MS350 GigE AP ports provide 1G wired uplinks.
CLIENT TESTS	Verify authentication, DHCP, DNS, application reachability, roaming and concurrent-client performance.

No coverage guarantee or AP allocation per area is implied. RADIUS / identity services, floor dimensions and client performance targets require customer input.

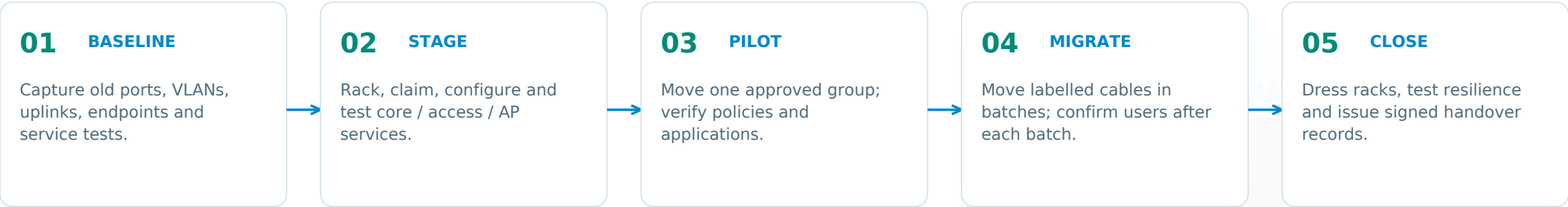
Dashboard build | From device inventory to live policies

Step	Configuration action	Required evidence
01 OWNERSHIP	Create / use customer-owned organisation; grant named administrator roles and enforce approved MFA.	Organisation owner, authorised roles and handover access list.
02 INVENTORY	Claim serials; assign networks; name devices by site / rack / role and label physical hardware.	8 switches + 35 APs = 43 ordered managed devices; serial register.
03 STAGE	Provide management connectivity; validate DNS / gateway and current Dashboard-required firewall destinations.	All devices online; licence / firmware status recorded.
04 CORE STACK	Align supported firmware; assemble core data-stack ring and verify stack members. Configure Power-Stack per power design.	Stack health, member serials, power topology / budget and recovery test.
05 NETWORK	Apply VLANs, gateway / routes, STP and port profiles. Configure supported cross-stack LACP only after both ends match.	Port export, VLAN / IP map, link aggregation state and path checks.
06 WIRELESS	Map SSIDs, authentication and access rules; apply RF profiles and approved AP names / locations.	SSID policy, test clients and post-installation RF record.

Dashboard menu names vary by release. Use the supported current workflow; this is a configuration checklist, not an invented IOS command listing.
 The PO includes three-year licence lines; obtain the current licence state and supported firmware / lifecycle status before a new rollout or renewal.

Implementation | Switch replacement and controlled cutover

Before removing a live switch, capture its configuration and every connected cable. Stage the replacement and prove one representative access group first. Migrate in an approved maintenance window with application owners available for validation.



Work package	Field execution / rollback control
RACK / FIBRE	Check earthing, airflow, power feeds, fibre loss and polarity; label both ends and dress without bend-radius violations.
CUTOVER	Disconnect old links in the agreed sequence; patch only the validated replacement topology and monitor loops / reachability.
ROLLBACK	If a critical service fails: isolate new affected links, restore labelled old patching / configuration, then retest with the service owner.

Do not run disruptive cable diagnostics or reboot / failover tests on production links without the approved test window. Record before / after results.

Acceptance, operations and customer handover

Acceptance test	Expected result / customer record
CORE / FIBRE	Stack members healthy; intended uplinks negotiated at 10G; LACP / STP state verified; no unexpected loops or port errors.
USER / WI-FI	Representative wired and wireless clients obtain addresses, authenticate and reach approved services; guest internal access is blocked.
RESILIENCE	In an approved window, test one uplink / stack path failure and recovery; record measured interruption against agreed criteria.
CLOSEOUT	Serial inventory, cable / port map, VLAN / IP plan, RF report, licence status, Dashboard roles and signed test sheet.

Recommended operations: review offline devices, link / PoE alarms and client failures daily; review capacity, firmware, licences and access rights on an approved schedule.

CUSTOMER VALUE

One management view, standard port profiles and a documented fibre backbone support faster diagnosis and controlled changes. These are design benefits, not measured ROI.

CUSTOMER OWNERSHIP

Netafim approves identity, policies, outage windows and acceptance. D-TECH implements, tests, documents and transfers operational knowledge.

D-TECH SUPPORT

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